Question Bank PYQs

Chemical Equations and Reactions

2 Mark Questions

Q1: What is a chemical reaction? (2 Marks)

- A chemical reaction is a process in which one or more substances, the reactants, are converted to one or more different substances, the products.
- It involves the breaking and making of bonds between atoms.
- The substances produced have entirely different properties from the original substances.
- Example: Burning of magnesium in air to form magnesium oxide.

Q2: Define a balanced chemical equation. (2 Marks)

- A balanced chemical equation is one where the number of atoms of each element is the same on both the reactant and product sides.
- It follows the Law of Conservation of Mass.
- Mass can neither be created nor destroyed in a chemical reaction.
- Example: $2H_2 + O_2 \rightarrow 2H_2O$.

Q3: Why is respiration considered an exothermic reaction? (2 Marks)

- During respiration, glucose combines with oxygen in the cells of our body.
- This process breaks down glucose to produce energy, carbon dioxide, and water.
- Since energy is released in the form of heat/ATP, it is an exothermic process.
- Equation: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + Energy$.

Q4: What is a combination reaction? Give an example. (2 Marks)

- A reaction in which a single product is formed from two or more reactants is called a combination reaction.
- It generally involves the release of energy.
- Example: Burning of coal ($C + O_2 \rightarrow CO_2$).

- Example: Formation of water from hydrogen and oxygen.

Q5: Define a decomposition reaction. (2 Marks)

- A reaction in which a single reactant breaks down to give simpler products is called a decomposition reaction.
- It is the opposite of a combination reaction.
- It usually requires energy in the form of heat, light, or electricity.
- Example: $CaCO_3 \xrightarrow{Heat} CaO + CO_2$.

Q6: What do you mean by a precipitation reaction? (2 Marks)

- Any reaction that produces an insoluble solid (precipitate) is called a precipitation reaction.
- The precipitate settles at the bottom of the solution.
- These are usually double displacement reactions.
- Example: Mixing sodium sulphate and barium chloride solutions.

Q7: Define oxidation and reduction in terms of oxygen gain/loss. (2 Marks)

- Oxidation: The gain of oxygen by a substance during a reaction.
- Reduction: The loss of oxygen by a substance during a reaction.
- If a substance gains oxygen, it is said to be oxidized.
- If a substance loses oxygen, it is said to be reduced.

Q8: What are antioxidants? Why are they added to fat-containing foods? (2 Marks)

- Antioxidants are substances which prevent oxidation.
- They are added to foods containing fats and oils to prevent them from becoming rancid.
- They slow down the process of oxidation.
- This helps in preserving the smell and taste of the food.

Q9: What is a displacement reaction? (2 Marks)

- A reaction in which a more reactive element displaces a less reactive element from its compound.

- It depends on the reactivity series of metals.
- Example: Iron displacing copper from copper sulphate solution.
- Equation: $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$.

Q10: Why do we apply paint on iron articles? (2 Marks)

- Paint is applied to prevent iron articles from rusting.
- It creates a barrier between the iron surface and the atmosphere.
- It prevents the iron from coming into contact with moisture and oxygen.
- This inhibits the chemical reaction that causes corrosion.

3 Mark Questions

Q11: Why should a magnesium ribbon be cleaned before burning in air? (3 Marks)

- Magnesium is a very reactive metal.
- When stored, it reacts with oxygen in the air to form a stable layer of magnesium oxide on its surface.
- This oxide layer is unreactive and prevents the underlying magnesium from burning.
- Cleaning the ribbon with sandpaper removes this protective layer.
- This allows the metal to burn efficiently and produce a dazzling white flame.

Q12: Identify the substances that are oxidized and the substances that are reduced in the following reaction: $CuO + H_2 \rightarrow Cu + H_2O$ (3 Marks)

- In this reaction, Copper Oxide (CuO) is losing oxygen to become Copper (Cu).
- Therefore, CuO is being reduced.
- Hydrogen (H_2) is gaining oxygen to form water (H_2O).
- Therefore, H_2 is being oxidized.
- This is a classic example of a Redox reaction where oxidation and reduction occur simultaneously.

Q13: Why is the amount of gas collected in one of the test tubes in the electrolysis of water double of the amount collected in the other? (3 Marks)

- Water (H_2O) is composed of two parts of hydrogen and one part of oxygen by volume.
- The chemical equation for electrolysis is: $2H_2O \xrightarrow{Electricity} 2H_2 + O_2$.
- According to the stoichiometry of the reaction, 2 moles of Hydrogen gas are produced for every 1 mole of Oxygen gas.
- Therefore, the volume of hydrogen gas collected is double the volume of oxygen gas.
- Hydrogen is collected at the cathode, while oxygen is collected at the anode.

Q14: What happens when dilute hydrochloric acid is added to iron filings? Write the equation. (3 Marks)

- When dilute HCl is added to iron filings, a displacement reaction occurs.
- Hydrogen gas and iron (II) chloride are produced.
- The reaction is: $Fe + 2HCl \rightarrow FeCl_2 + H_2$.
- Bubbles of hydrogen gas are evolved, which can be tested with a burning matchstick (it burns with a pop sound).
- The iron filings gradually dissolve as they react to form the salt.

Q15: Why does the color of copper sulphate solution change when an iron nail is dipped in it? (3 Marks)

- When an iron nail is dipped in copper sulphate ($CuSO_4$) solution, a displacement reaction takes place.
- Iron is more reactive than copper, so it displaces copper from the solution.
- The blue color of the copper sulphate solution fades and turns light green due to the formation of ferrous sulphate ($FeSO_4$).
- A reddish-brown coating of copper metal is deposited on the iron nail.
- Equation: $Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$.

Q16: Explain decomposition reactions with one example each for heat, light, and electricity. (3 Marks)

- Thermal Decomposition (Heat): $2FeSO_4 \xrightarrow{Heat} Fe_2O_3 + SO_2 + SO_3$.
- Electrolytic Decomposition (Electricity): $2H_2O \xrightarrow{Electricity} 2H_2 + O_2$.
- Photolytic Decomposition (Light): $2AgCl \xrightarrow{Sunlight} 2Ag + Cl_2$.
- In all these cases, a single reactant breaks down into two or more products using external energy.

Q17: What is a double displacement reaction? Explain with the help of an example. (3 Marks)

- A reaction in which there is an exchange of ions between the reactants to form new compounds is called a double displacement reaction.
- Example: Reaction between sodium sulphate and barium chloride.
- Equation: $Na_2SO_4(aq) + BaCl_2(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$.
- In this reaction, Ba^{2+} and Na^+ ions exchange places.
- A white precipitate of barium sulphate ($BaSO_4$) is formed.

Q18: What is rancidity? List two ways to prevent it. (3 Marks)

- Rancidity is the process of slow oxidation of fats and oils present in food materials, resulting in a change in smell and taste.
- It makes the food unpleasant and unfit for consumption.
- Prevention 1: Adding antioxidants to food (e.g., BHA or BHT).
- Prevention 2: Keeping food in airtight containers to reduce oxygen exposure.
- Prevention 3: Flushing food packets (like chips) with nitrogen gas to create an inert atmosphere.

Q19: Explain the observation when lead nitrate powder is heated in a boiling tube. (3 Marks)

- When lead nitrate $[Pb(NO_3)_2]$ is heated, it undergoes thermal decomposition.
- The observation includes the emission of brown fumes of Nitrogen dioxide (NO_2).
- A yellow residue of Lead oxide (PbO) is left behind in the test tube.
- Equation: $2Pb(NO_3)_2 \xrightarrow{Heat} 2PbO + 4NO_2 + O_2$.

- Oxygen gas is also released during this reaction.

Q20: Why is it important to balance a chemical equation? (3 Marks)

- According to the Law of Conservation of Mass, mass can neither be created nor destroyed in a chemical reaction.
- The total mass of elements present in the products must be equal to the total mass of elements present in the reactants.
- The number of atoms of each element remains the same before and after a chemical reaction.
- Balancing ensures that the equation accurately represents the quantitative relationships between reactants and products.

5 Mark Questions

Q21: Classify the types of chemical reactions with a definition and one balanced equation for each. (5 Marks)

- Combination Reaction: Two or more reactants combine to form a single product. Example: $CaO + H_2O \rightarrow Ca(OH)_2 + Heat$.
- Decomposition Reaction: A single reactant breaks down into two or more simpler products. Example: $2Pb(NO_3)_2 \xrightarrow{Heat} 2PbO + 4NO_2 + O_2$.
- Displacement Reaction: A more reactive element displaces a less reactive element from its salt solution. Example: $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$.
- Double Displacement Reaction: Two compounds react by an exchange of ions to form two new compounds. Example: $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$.
- Redox Reaction: A reaction where both oxidation and reduction occur simultaneously. Example: $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$.

Q22: Describe an activity to demonstrate the electrolysis of water. Mention the observations and the conclusion. (5 Marks)

- Procedure: Take a plastic mug, drill two holes at its base, and fit rubber stoppers. Insert carbon electrodes and connect them to a 6V battery. Fill the mug with water and add a few drops of dilute sulphuric acid to make it conducting.

- Invert two test tubes filled with water over the two carbon electrodes.
- Observation 1: Bubbles of gas are formed at both electrodes, displacing water in the test tubes.
- Observation 2: The volume of gas collected at the cathode (negative electrode) is double the volume of gas collected at the anode (positive electrode).
- Test: The gas at the cathode burns with a pop sound (Hydrogen). The gas at the anode supports combustion (Oxygen).
- Conclusion: Water decomposes into Hydrogen and Oxygen in a 2:1 ratio by volume when electricity is passed through it.

Q23: Explain the process of corrosion of iron (rusting). What are the necessary conditions and how can it be prevented? (5 Marks)

- Definition: Corrosion is the process in which metals are eaten up gradually by the action of air, moisture, or chemicals on their surface.
- Rusting of Iron: When iron is exposed to moist air for a long time, it acquires a coating of a brown flaky substance called rust ($Fe_2O_3 \cdot xH_2O$).
- Conditions Necessary: (i) Presence of air (Oxygen), (ii) Presence of moisture (Water).
- Prevention - Barrier Method: Painting, oiling, or greasing the metal surface to prevent contact with air/moisture.
- Prevention - Galvanization: Coating iron with a thin layer of zinc. Zinc is more reactive and corrodes itself to protect the iron.
- Prevention - Alloying: Mixing iron with other metals/non-metals (e.g., Stainless steel) to make it resistant to corrosion.

Q24: Explain the thermal decomposition of Ferrous Sulphate crystals with the help of observations and a chemical equation. (5 Marks)

- Initial State: Ferrous sulphate crystals ($FeSO_4 \cdot 7H_2O$) are green in color.
- Heating: On heating, the crystals first lose water of crystallization and turn white.
- Chemical Change: Further heating causes the anhydrous ferrous sulphate to decompose into ferric oxide (Fe_2O_3), sulphur dioxide (SO_2), and sulphur trioxide (SO_3).

- Equation: $2FeSO_4(s) \xrightarrow{Heat} Fe_2O_3(s) + SO_2(g) + SO_3(g)$.
- Observations: (i) Change in color from green to reddish-brown (due to Fe_2O_3), (ii) Characteristic smell of burning sulphur due to SO_2 and SO_3 gases.
- Type: This is a thermal decomposition reaction because a single reactant breaks down into three products using heat.

Q25: What are Redox reactions? Identify the oxidizing agent, reducing agent, substance oxidized, and substance reduced in the reaction of Manganese dioxide with Hydrochloric acid. (5 Marks)

- Definition: Redox reactions (Reduction-Oxidation) are chemical reactions in which one reactant gets oxidized while the other gets reduced.
- Reaction Equation: $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$.
- Substance Oxidized: HCl is oxidized to Cl_2 because it loses hydrogen.
- Substance Reduced: MnO_2 is reduced to $MnCl_2$ because it loses oxygen.
- Oxidizing Agent: MnO_2 (it provides oxygen or accepts hydrogen to oxidize HCl).
- Reducing Agent: HCl (it provides hydrogen or removes oxygen to reduce MnO_2).
- In summary, the oxidizing agent is always the substance being reduced, and the reducing agent is the substance being oxidized.

Q26: Differentiate between Exothermic and Endothermic reactions. Provide two examples for each from everyday life. (5 Marks)

- Exothermic Reactions: Reactions in which heat is released along with the formation of products. The temperature of the surroundings rises.
- Endothermic Reactions: Reactions in which energy is absorbed from the surroundings in the form of heat, light, or electricity to proceed.
- Exothermic Example 1: Respiration ($C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + Energy$).
- Exothermic Example 2: Burning of natural gas ($CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + Heat$).
- Endothermic Example 1: Photosynthesis ($6CO_2 + 6H_2O \xrightarrow{Sunlight} C_6H_{12}O_6 + 6O_2$).
- Endothermic Example 2: Decomposition of Calcium Carbonate ($CaCO_3 \xrightarrow{Heat} CaO + CO_2$).

Q27: Explain the displacement reaction of an iron nail in copper sulphate solution. Why is it classified as a chemical change? (5 Marks)

- Experimental Setup: Take two iron nails and clean them with sandpaper. Dip one nail in a test tube containing blue-colored copper sulphate solution for 20 minutes.
- Observation (Color): The blue color of the copper sulphate solution fades and changes to a light green color.
- Observation (Nail): The iron nail becomes brownish in color due to the deposition of copper metal.
- Chemical Equation: $Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$.
- Reasoning: Iron, being more reactive than copper, displaces copper from its salt solution to form ferrous sulphate.
- Classification: It is a chemical change because a new substance (ferrous sulphate) is formed with entirely different properties, and the process is not easily reversible.

Q28: Discuss the significance of the photolytic decomposition of silver chloride and silver bromide. How is it used in technology? (5 Marks)

- Process: Silver chloride ($AgCl$) and Silver bromide ($AgBr$) are sensitive to light.
- Decomposition of $AgCl$: When white silver chloride is kept in sunlight, it turns grey due to the formation of silver metal and chlorine gas. $2AgCl(s) \xrightarrow{Sunlight} 2Ag(s) + Cl_2(g)$.
- Decomposition of $AgBr$: Similarly, silver bromide decomposes into silver and bromine. $2AgBr(s) \xrightarrow{Sunlight} 2Ag(s) + Br_2(g)$.
- Observation: The change in color from white/yellow to grey is a visual indicator of the chemical reaction.
- Technological Application: These reactions were traditionally used in black and white photography.
- Storage: Because they are highly light-sensitive, these chemicals are stored in dark-colored bottles to prevent accidental decomposition.

Q29: Describe the reaction between Quick Lime (CaO) and Water. What are the characteristics of this reaction? (5 Marks)

- Reaction: Calcium oxide (Quick Lime) reacts vigorously with water to produce Calcium hydroxide (Slaked Lime).
- Equation: $CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq) + Heat$.
- Characteristic 1 (Combination): Two substances combine to form a single product ($Ca(OH)_2$).
- Characteristic 2 (Exothermic): A large amount of heat is evolved during the reaction, causing the container to become very hot.
- Characteristic 3 (Observation): A hissing sound is often heard as the reaction occurs.
- Application: The slaked lime produced is used for white-washing walls. It reacts slowly with CO_2 in air to form a thin, shiny layer of $CaCO_3$ on the walls.

Q30: What are the various observations that help us determine whether a chemical reaction has taken place? Provide one example for each. (5 Marks)

- Change in State: Some reactions involve a change from solid to liquid or gas. Example: Combustion of wax (solid to liquid/gas).
- Change in Color: The color of reactants changes after the reaction. Example: Iron nail in copper sulphate (Blue to Green).
- Evolution of Gas: Many reactions produce gas bubbles. Example: Zinc granules reacting with dilute sulphuric acid produces Hydrogen gas ($Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$).
- Change in Temperature: The reaction may be exothermic (releases heat) or endothermic (absorbs heat). Example: Reaction of quick lime with water.
- Formation of a Precipitate: An insoluble solid is formed. Example: Reaction between Lead Nitrate and Potassium Iodide forms a yellow precipitate of Lead Iodide.
- By observing any of these changes, we can confirm the occurrence of a chemical reaction.